10.6 Solving Problems Involving Inscribed Angles

Lesson Introduction

Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.	Learning Target	• I can use the relationship between the measure of an inscribed angle and its intercepted arc to solve problems.

Benchmark Coherence	**Building On**	**Working Toward**
	N/A	N/A

Essential Question	• What is the relationship between an inscribed angle and the measure of the intercepted arc?
Lesson Narrative	In this lesson, students are introduced to the concept of inscribed angles. Students will begin by exploring the concepts of inscribed angles to discover the inscribed angles theorem. They will then apply that knowledge to solving mathematical problems that involve inscribed angles using algebraic expressions.

Component Summary	10.6.1 Warm-Up (~5 min.)	Determine measures of angles on a clock *(SE p. 148)*	10.6.3 Guided Practice (15 min.)	Solve problems using inscribed angles *(SE p. 150)*
	10.6.2 Exploration (10-15 min.)	Discover the relationship between an inscribed angle and its intercepted arc *(SE p. 149)*	10.6.4 Your Turn! (10 min.)	Solve problems using inscribed angles *(SE p. 152)*
	10.6.5 Wrap-Up (~5 min.)	Reflect on what is still confusing from the lesson *(SE p. 153)*		

Resources	**Glossary**	**Materials**	**Additional Preparation**
	Key Terms: • Chord • Inscribed angle • Inscribed angle theorem Additional Terms: • Central angle • Arc measure • Intercepted arc	• (Optional) Chart paper • (Optional) Markers • (Optional) GeoGebra • (Optional) Protractor	• N/A

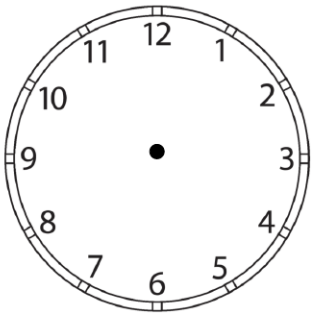
Teacher Tips	Common Misconceptions • Students may think the inscribed angle is equal to the intercepted arc.	Things to Consider • Do not give away anything too early in 10.6.2 Exploration, as it is designed to be a discovery activity where students discover the inscribed angle theorem.
	Literacy and Language Routines **Think-Pair-Share** During 10.6.2 Exploration, students will have independent time to think through and complete one section of the activity (a-c), and then collaborate with their partner in part d to compare conclusions before moving to the final parts of the activity.	Mathematical Thinking and Reasoning (MTR) Standard **MA.K12.MTR.5.1: Patterns and Structure** For 10.6.3 Guided Practice and 10.6.4 Your Turn!, students will be applying what they learned in the prior activity about inscribed angles to solve mathematical problems that involve algebraic expressions. To do this, students will need to focus on the relevant details within the problem, and decompose it into manageable parts.
Differentiate Instruction	To increase student engagement, consider making parts of this lesson digital. For 10.6.2 Exploration, consider supporting the activity using the "Central Angles," "Inscribed Angles," and the "Intercepted Arcs" activity from GeoGebra or other digital resources.	
Lesson Notes:		

10.6.1 Warm-Up: Angles on a Clock

Component Overview: *Students will explore the angles that are created by three different times of a standard clock.*

10.6.1 Warm-Up: Angles on a Clock

1. Use the clock to answer the following questions.



a. If the time is $10{:}10$ and the hour hand does not move, what is the exact angle between the minute hand and hour hand?
$120°$ *or* $240°$

b. If the hour hand does move, what is the exact angle if the time is $7{:}30$?
$45°$ *or* $315°$

c. If the hour hand does move, what is the exact angle if the time is $7{:}35$?
$17.5°$ *or* $342.5°$

For this activity, consider providing students with a protractor to help them verify that their angle measures are correct after trying without one.

Some students will struggle with reading a standard clock. Be prepared to provide images of what $10{:}10$ and $7{:}35$ represent.

10.6.2 Exploration: Inscribed Angles

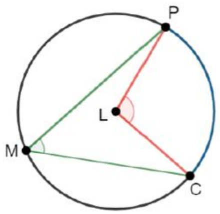
Component Overview: *Students will explore the concept of inscribed angles through a series of two, multi-part, scaffolded questions. For this activity, students will start independently before sharing answers with a partner.*

10.6.2 Exploration: Inscribed Angles

1. Circle L is shown where P, C, and M are points on the circle.

A ***chord*** is a line segment on the interior of a circle with both endpoints lying on the circle.

An ***inscribed angle in a circle*** is an angle which is formed in the interior of a circle when two chords share an endpoint.



Central Angle $m\angle PLC = 110°$

Inscribed Angle $m\angle PMC = 55°$

Arc Measure $\overparen{PC} = 110°$

a. What is the connection between the central angle and the arc measure?
The measure of a central angle is equal to the measure of its intercepted arc.

b. Describe the relationship between the central angle and the inscribed angle.
The measure of an inscribed angle is half that of its intercepted central arc.

c. Write a conclusion about the measures of the central angle, the inscribed angle, and the arc measure.
Answers will vary. For an inscribed angle, a central angle, and an arc that all intercept the same two points on a circle the central angle and the arc have equal measures while the measure of the inscribed angle is half that of the central angle.

Consider creating a "parts of a circle" anchor chart where students can see all the different components with circles in one place. The chart should include radius, diameter, chord, arc, central angle, and inscribed angle. Only include vocabulary on this chart so that students can visualize the components, not all the relationships associated with each.

10.6.2 Exploration: Inscribed Angles (continued)

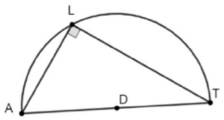
d. Share your response to Part C with two classmates. Listen to their conclusion and summarize it in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating your summary was correct.

My Summary of Their Conclusion	Initials
Answers will vary.	

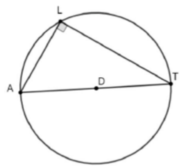
e. Using Circle L, complete the table.

Central Angle	Inscribed Angle	Arc Measure
$216°$	$108°$	$216°$
$162°$	$81°$	$162°$
$143°$	$71.5°$	$143°$
$290°$	$145°$	$290°$

2. $\angle ALT$ is inscribed in semicircle D, as shown.



a. Describe the angle inscribed in the semicircle.
$\angle ALT$ is a right angle



b. If the entire Circle D is shown, what is the measure of $\widehat{AT}$?
$m\,\widehat{AT} = 180°$

c. Discuss with your partner the relationship between an inscribed angle and a semicircle.
An inscribed angle that intercepts the endpoints of a semicircle is a right angle.

Inscribed Angle Theorem
The measure of an inscribed angle is one-half the measure of the intercepted arc.

When circulating during this activity, listen for student responses to question 1 part d. This is a place where misconceptions, such as the relationship of the angle being the same instead of half, can be heard and addressed.

Consider asking the following for students stuck on question 1 part e:
- *What do you know about the relationship between the central angle and the arc measure from the prior lesson? Consider filling in those answers first.*
- *What did you notice about the relationship between the inscribed angle and the arc measure?*

Consider asking the following with question 2 part b:
- *Is the arc created by the endpoints of a diameter a major arc or a minor arc? Explain your answer.*
- *How do we know which arc is being referred to when naming an arc created by a semi-circle?*

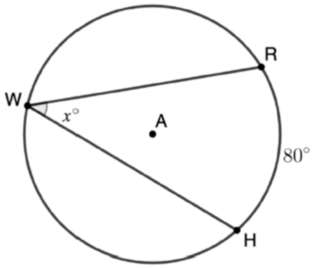
Students may get stumped using the "semicircle" in question 2 part c. When circulating, remind students who are struggling to describe the relationship between a semicircle and a circle, and how that relationship impacts that inscribed angle.

10.6.3 Guided Practice: Inscribed Angles

Component Overview: *Students will complete four practice problems finding the missing measures and/or values using what they have learned about inscribed angles.*

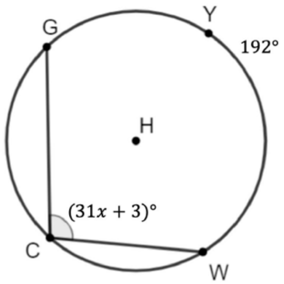
10.6.3 Guided Practice: Inscribed Angles

1. Circle A is shown where points R, W, and H are on the circle. $\angle RWH$ is inscribed inside Circle A with $m\,\widehat{RH} = 80°$. Find the value of x.
   $x = 40$



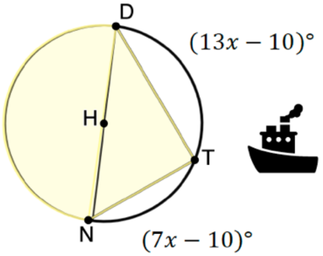
For this activity, consider beginning with a teacher-modeled activity for question 1. In this activity, students will be applying what they have learned during the Exploration. Students can benefit from hearing structured thinking out loud on determining what is known from the problem, what relationships can be determined, and how those relationships can be used to solve problems.

2. Points Y, G, C, and W lie on Circle H. $\angle GCW$ is inscribed in the circle. Find x, given $m\,\widehat{GYW} = 192°$ and $m\angle GCW = (31x + 3)°$.
   $x = 3$



Students may struggle creating a diagram of the circle in question 2. Consider providing a circle with the points and angle drawn but with no labels as an entry point for those students into the problem.

3. Boats use angle measures and arcs when sailing in treacherous waters. When a boat captain turns on the boat light, it forms inscribed angle $\angle NTD$. Use the information for Circle H to find $m\angle NDT$.

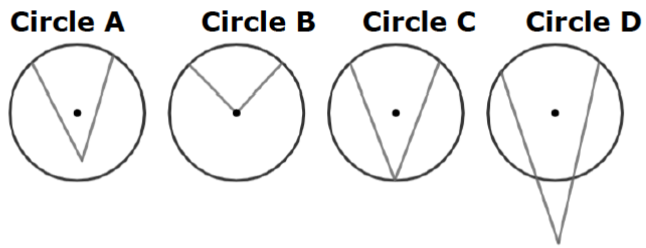


$m\angle NDT = 30°$

Question 3 is the first real-world example with this content for this lesson. Some students may need assistance with pulling information from the problem, because they could be intimidated by the word problem.

10.6.3 Guided Practice: Inscribed Angles (continued)

4. Craig and Valentina are reviewing circles to identify inscribed angles.



Valentina identified Circle C is an inscribed angle. Craig identified both Circle A and Circle C as inscribed angles.

Do you agree with Valentina, Craig, or neither? Justify your reasoning.
Answers will vary. I agree with Valentina because inscribed angles are formed by two chords that intersect at the same point on the circumference of the circle.

Question 4 exhibits key features of MA.K12.MTR.4.1: Discussion and Reflection by having students analyze the mathematical thinking of others.

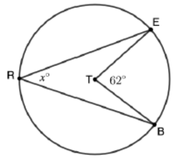
If students finish early, challenge them with writing a conjecture about the relationship of the angle in the image of Circle D to the arcs of the circle. Have them include a summary of their thinking.

10.6.4 Your Turn!

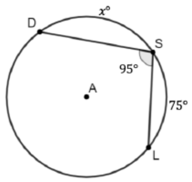
Component Overview: *Students will complete two independent practice problems with solving for angle measures and/or values using the inscribed angle relationships.*

10.6.4 Your Turn!

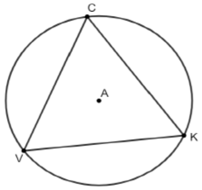
1. Circle T is shown where $\angle ETB$ is a central angle, $\angle ERB$ is inscribed in the circle, and $m\angle ETB = 62°$. Find $m\angle ERB$.
   $x = 31°$



2. Circle A has points D, S, and L on the circle. $\angle DSL$ is inscribed in Circle M measuring $95°$. Given $m\,\widehat{SL} = 75°$, find $m\,\widehat{DS}$.
   $m\,\widehat{DS} = 95°$



3. Equilateral triangle CKV is inscribed in Circle A. Circle A consists of three equal inscribed angles. Determine the value of the inscribed angles.
   $m\angle C = m\angle V = m\angle K = 60°$



This activity is designed to give students independent practice on what they have learned in this lesson. Student performance on this activity can provide valuable insight into which parts students may be struggling with.

Use these problems as an opportunity to work with small groups or individual students who are still struggling with the concept.

10.6.5 Wrap-Up: Reflection

Component Overview: *Students will identify the most difficult or confusing point in this lesson.*

10.6.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on solving problems using the relationship between the measure of an inscribed angle and its intercepted arc?

   Answers will vary. The muddiest point in today's lesson was remembering whether it was half the value or the same value. The muddiest point in today's lesson was remembering the definition of an inscribed angle.

Consider pairing students to discuss their reflections to see if peers can clear up any misconceptions of the lesson.